

JEPA World Model for Vision-Based Pick and Place

Project Proposal — revised per course feedback (22 Sept 2026)

Idea

We aim to build a world-model-based robot manipulation system. A simulated arm in MuJoCo picks a randomized object, carries it past randomized obstacles, and places it at a randomized target, observed through a static camera and a wrist camera.

We use a JEPA world model: a frozen, pretrained V-JEPA 2 encoder that turns images into latent representations, and a dynamics model that predicts how those representations evolve under candidate future actions (“imagination”). On top of this world model we compare two controllers: (1) a CEM/MPC planner that searches for good action sequences directly in the imagined rollouts, and (2) an actor-critic policy trained with RL on imagined rollouts. The target pose is encoded as a goal vector alongside the latent state, and the wrist camera adds a close-up view of contact and fine manipulation detail.

Pipeline overview:

- Episodes are collected in MuJoCo using scripted pick-and-place routines (including noise, random actions, and deliberate collisions), recording camera frames, actions, and rewards (e.g. reaching, grasping, object state, collision penalties).
- The frozen encoder converts images to latent representations. Two networks are trained on these latents: a dynamics model predicting the next latent state, and a reward model predicting the reward of a latent state (used for imagined rollouts).
- The first controller, a CEM/MPC planner, samples action sequences, scores them via the dynamics + reward model's imagined rollouts, and executes the best-scoring action.
- The second controller, an actor-critic policy, is trained on imagined rollouts and on newly collected real data.
- At deployment, collision avoidance is ideally learned directly from the JEPA model: camera images, robot state, and goal vector pass through the frozen encoder and trained policy at every control step.

Group and Work Split

Members: Kuba de Quattro, Carl Fredrik Nexmark, Sarvesh Raj Kumar

- **Member 1 (M1) — Environment & data:** MuJoCo task setup, data collection, evaluation
- **Member 2 (M2) — World model:** JEPA encoder integration, dynamics and reward model training
- **Member 3 (M3) — Control:** CEM/MPC planner, RL actor-critic, testing and tuning

Task Breakdown by Grade Level

Each level builds on the previous one — E, C, and A are cumulative, not separate scopes.

Grade E — baseline pipeline (Week 1–2)

Goal: minimal working pick-and-place loop, single object, no randomization.

- **M1:** MuJoCo task setup (fixed object + target), scripted episodes, initial data collection
- **M2:** V-JEPA 2 encoder integration — get it working; wire in a first, rough dynamics model (doesn't need to be accurate)
- **M3:** Simple CEM planner, tested against the rough dynamics model

Deliverable: arm attempts pick-and-place using the CEM planner on the JEPA world model — doesn't need to succeed reliably yet.

Grade C — randomized, two-controller pipeline (Week 2–3)

Goal: randomized pipeline, reward model added, both controllers running.

- **M1:** Add randomization (object, obstacles, target), expand dataset, add wrist-camera data
- **M2:** Validate the encoder's latents are actually useful, add wrist-camera fusion, train the dynamics model properly, train + validate the reward model
- **M3:** Upgrade to proper MPC, implement actor-critic, begin training on imagined rollouts

Deliverable: pick-and-place works reliably across randomized layouts; first CEM/MPC vs. RL-policy comparison.

Grade A — full comparison and generalization (Week 3–4)

Goal: full comparison and generalization results.

- **M1:** Policy-augmented data collection, generalization evaluation sweeps
- **M2:** Retrain and validate models on the expanded dataset
- **M3:** Complete actor-critic training, full CEM/MPC vs. RL comparison, evaluate generalization

Deliverable: full planner-vs-policy comparison with generalization results, written up for the final report.

Time Plan

Weeks map onto the grade levels above, so progress toward a higher grade is visible from week to week.

Week	Target	M1 – environment & data	M2 – world model	M3 – control
1	E	MuJoCo task setup + scripted episodes, initial data collection	Encoder integration; dynamics model trained on initial dataset	CEM planner + RL policy skeleton code
2	E → C	Add randomization; collect full dataset; evaluate as needed	Dynamics + reward model trained and validated	First full pick-and-place attempt with CEM planner
3	C → A	Collect new data using the trained policy	Retrain models as needed on expanded data	Evaluate CEM/MPC vs. RL policy
4	A (buffer)	Generalization evaluation across new objects/layouts	Final model validation	Final comparison + results write-up

Possible Risks

- V-JEPA 2 is pretrained on natural video, not synthetic renders — wiring it up correctly (clip length, resolution, normalization) is the project's biggest single integration risk, and there may be a domain gap that limits how informative the frozen latents are. Fallback: swap in a simpler frozen image encoder (e.g. a pretrained ResNet/CLIP) to unblock M2/M3 and revisit V-JEPA 2 once the rest of the pipeline is stable.
- Grasping random objects is hard to explore; data may not be sufficient or diverse enough for generalization.
- RL may underperform due to errors in the learned reward model; the CEM/MPC planner serves as a fallback controller.
- Data collection volume may create storage constraints.

Backup Plan

Scripted pick-and-place with exploration may prove too hard. In that case, we would swap the use case to a mobile robot navigating to a random goal pose among randomly spawned obstacles, using one fixed onboard camera. The use case and reward definitions would change, but the rest of the pipeline (JEPA encoder, dynamics/reward models, CEM/MPC vs. RL comparison, and the E/C/A task breakdown above) stays the same.